

# ALLAN ILYASOV

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## SUMMARY

Data Science graduate from St. John's University with hands-on experience building and deploying AI-powered applications. Skilled in machine learning, full-stack development, and cloud infrastructure, with a track record of taking projects from idea to production. Passionate about using software and AI to tackle problems that benefit society, from making education more accessible to improving how people interact with technology.

## EXPERIENCE

### **Graduate Research Assistant – St. John's University**

**September 2025 – May 2026**

#### **Codify AI – Advancing Education Through Generative AI & LLM (Dr. Fazel Keshtkar):**

- Product manager & lead architect of [codifyai.org](https://codifyai.org), an AI-powered programming tutor using LangChain agents, Claude Haiku 4.5 via AWS Bedrock, and RAG for streaming responses, and context-aware tutoring in Java, Python & SQL.
- Designed production-ready AWS infrastructure (Elastic Beanstalk, DynamoDB, Route 53, HTTPS) with Docker containerization, supporting scalable REST API architecture.
- Built React-based adaptive learning platform with persistent state management, real-time progress tracking.
- Earned “Commercializable Project” award at the 2025 St. John's University's Student Research Conference.
- Published research in the FLAIRS-39 (2026) proceedings. [doi.org/10.32473/flairs.39.1.141554](https://doi.org/10.32473/flairs.39.1.141554)

### **Manager – Uncle Edik's Pickles**

**December 2019 – Present**

- Grew a home-based pickle start-up and supported its growth into a brick-and-mortar factory and national brand.
- Trained & led a team of 5 employees in production of thousands of jars per week of 9 different pickled products.
- Engineered Apps Script automations in Google Sheets to streamline wholesale invoicing and business analytics.
- Managed sales to customers, wholesale delivery to supermarkets, and inventory replenishment from Restaurant Depot.
- Designed branded merchandise, and produced viral marketing content, generating millions of views.
- Ensure compliance with NYS Food Safety regulations & maintained quality control with daily production checklists.

## SELECT PROJECTS

### **TaxSnap – Receipt Management App | [github.com/allanilya/TaxSnap](https://github.com/allanilya/TaxSnap)**

- Built an iOS app letting users scan receipts offline, auto-extracting merchant, date, and amounts with on-device OCR.
- Integrated OpenAI embeddings with KNN classification for automatic receipt categorization across 100+ merchants.
- Implemented Core Data + CloudKit sync and multi-format export (CSV, Excel, PDF) for IRS-compliant tax records.

### **Time Series Forecasting for Financial Markets | [github.com/allanilya/Stock-analytics](https://github.com/allanilya/Stock-analytics)**

- Created ARIMA, LSTM, & GRU forecasting models across AAPL, NVDA, & LYFT, reducing RMSE by up to 82%.
- Implemented grid search across 36 ARIMA configurations and built 2-layer LSTM/GRU architectures with dropout regularization utilizing Python, TensorFlow, and Statsmodels.

### **Diabetes Prediction Model | [github.com/allanilya/Diabetes-Prediction](https://github.com/allanilya/Diabetes-Prediction)**

- Trained Random Forest, Logistic Regression & KNN models with 5-fold cross-validation to predict diabetes risk.
- Investigated lifestyle and demographic predictors of diabetes using 230K+ patient records.
- Prepared data through imputation, encoding, outlier detection and addressed class imbalance leveraging SMOTE.

## EDUCATION

### **St. John's University – Queens, New York**

#### **Master of Science, Data Science**

**May 2026**

#### **Bachelor of Science, Computer Science**

**May 2025**

- Dean's List 2021-2025 | Member of the Association for Computing Machinery | EPIC Mentoring Program, Student Mentor

## SKILLS SUMMARY

- **Programming Languages:** Python, Java, SQL, R, JavaScript, Swift, TypeScript, C++, PHP
- **Frameworks & Libraries:** React, Next.js, Django, Flask, Spring Boot, Node.js, Bootstrap, LangChain, LangGraph, REST APIs, Pandas, NumPy, Scikit-learn, TensorFlow/Keras, PyTorch, Spark, Beautiful Soup, NLTK, Spacy
- **Cloud & DevOps:** AWS, Docker, DynamoDB, Elastic Beanstalk, Lambda, API Gateway
- **Linguistic Skills:** Russian native speaker | Intermediate proficiency in Spanish

**Work Eligibility:** US Citizen